

David Geddam

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EDUCATION

Bob Jones University

Bachelor of Science in Computer Science

Greenville, SC

Expected graduation: May 2027

RESEARCH

Fault Prediction in Wind Turbines via Joint-Embedding Predictive Architecture

Jan. 2026

Developing · github.com/dave21-py/Predictive-maintenance-project (code)

- Building a Self-Supervised World Model that learns latent dynamics of 52-dim. sensor data from 95 wind turbines across 3 wind farms, grouping sensors into Controls, Environment, and State subsystems.
- Implementing VICReg loss to prevent representation collapse during self-supervised training, with Transformer-based encoder learning causal relationships between physical subsystems.

EXPERIENCE

Bob Jones University

Student Software Developer

Greenville, SC

02/2026 – present

- Building and maintaining web applications.

Independent Research

Research Assistant

New York City Metropolitan Area

12/2025 – present

- Supervisor: Sumanth Ratna. Kandavalli (ratna@nyu.edu).

Lovable

Campus Leader

Remote

01/2026 – present

- Represent AI software development platform, managing partnerships, and organizing AI-assisted development hackathons.

PROJECTS

Tiny Siri - Edge-Optimized Intent Classification | *Python, PyTorch*

December 2025

- Built a voice intent classifier by fine-tuning DistilBERT with a custom data augmentation pipeline, achieving 97 percent test accuracy and deploying it via Hugging Face Spaces and Streamlit. Optimized the model for on-device deployment using PyTorch dynamic quantization, reducing memory footprint by 48 percent (255MB → 132MB) while maintaining partial precision.

Student Performance Prediction | *Python, Flask, XGBoost*

November 2025

- Built end-to-end ML pipeline predicting student test scores, implementing modular codebase structure with custom exception handling, OOD, and CI/CD deployment. Evaluated 6+ regression models (Linear, Ridge, XGBoost, CatBoost), using cross-validation, and deployed best performer via Flask web app.

Medical diagnosis with Hybrid Vision Transformers | *Python, PyTorch*

October 2025

- Built a deep learning pipeline using a hybrid MobileNetV2 + Transformer architecture with self-attention mechanisms to classify brain dementia types and lung cancer malignancies, achieving 98 percent and 95 percent test accuracy respectively. Also used the Transformer's spatial feature processing to identify localized malignant patterns, with Grad-CAM heatmaps providing explainability.

Adversarial Vision Attacks | *Python, PyTorch*

September 2025

- This project demonstrates the fragility of modern Deep Learning Computer Vision systems (specifically ResNet-50) against Adversarial Attacks. By exploiting the gradients of the model, I successfully "poisoned" images that look identical to humans but are completely misclassified by the AI.

SKILLS

Programming: Python, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Git, Cursor